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October 20th, 2026

Team Charter

Github Link: https://github.com/elwoodruff/NowellWernerWoodruff.git

1. Team Purpose or Mission

What effects impact the size and duration of wildfires in California, particularly focusing on wildfire cause and prescribed burns over time and space?

Our goal is to explore and identify factors most significant in fire scale, spread, and impact. Our intended audience is people in California, nonacademics, and people generally impacted by fire risk. We all have Python and GIS skills, as well as a general appreciation for fire safety and impact. Our audience can expect high-quality, informative, and engaging data representation in the form of a dashboard.

2. Communication Plan and Collaborative Coding

We will communicate primarily via text/groupchat and use Google Docs to share documents and keep notes on meetings and information documented there. We will plan to respond within 24 hours to all group communication. In addition to the weekly meeting time, we will plan to meet weekly on Mondays at 4 pm for an hour in person.

3. Roles and Responsibilities

We are all interested to apply our Python, data analysis, and geospatial skills to a topic to a pertinent environmental and social issue. We plan to build on our geospatial experience through a Python lens, and create a product that implements our knowledge from multiple classes as well as learn new relevant skills.

We all plan to be involved in every relevant step. Before starting each step, we will plan based on people's schedules and level of effort from individuals in the weeks before. We plan to review work completed in the week before at each weekly meeting as well as assign work for the next week.

Specifically, Nick will be responsible for communicating with John, utilizing ArcGIS Pro for pulling from the geodatabase as necessary, and Python use in spatial analysis as necessary. Gwynn is responsible for reviewing all code and managing merge conflicts when they arise. Ella is responsible for recording notes and submitting documentation and ensuring all deadlines are met.

4. Operating Norms or Ground Rules

We will keep good notes through every meeting and rely on our plan's structure. We will be open about scheduling constraints and issues with our work, and will communicate regularly about milestones reached. We will provide each other organized files and readable code as well as details about decision-making that occurred through the coding process. If someone anticipates missing a meeting or deadline, they should communicate that as early as possible so the other members can accommodate. We will divide work as necessary in our Thursday class time and we will combine and review work in our weekly Monday meetings prior to submitting.

5. Values and Behaviors
A good teammate comes to meetings having completed the work they said they would, or has communicated effectively about things that have come up so work can be redistributed accordingly. Most important is communication and a willingness to help the group meet milestones in a way that equally distributes work over the course of the project. It also means a willingness to give feedback to group members when you feel expectations aren't being met and holding each other accountable. Setting expectations now will give us something to go back to in the event of any issues with group participation. By documenting our meetings, we can look back at how work has been distributed and plan future work accordingly.

6. Resources and Constraints
We plan on utilizing past notebooks for examples of organization and coding, as well as responsible use of AI where necessary and appropriate. We anticipate using AI for coding within ArcGIS Pro for spatial analysis as well as for building the dashboard once we move beyond our current Python abilities, as well as for troubleshooting code where necessary. John Fay can support coding work with issues with Python. The timeline, coding capabilities, and the available data will shape the scope of our project. We can put our contact information on the dashboard and clearly communicate the purpose of the process, and update the dashboard as necessary.

7. Decision-Making Guidelines
We will try to make most project decisions either in class time or during our meeting time. In the event of a necessary decision outside of scheduled meetings, we will try to plan another in person opportunity or ensure all group members agree before moving forward with a decision. For this reason, we will try to complete most work at a reasonable time of day and plan for deadlines. In the event of not being able to reach consensus, we will go with a majority, easy to do given that we have a group of three and seek alternative input from professors or teaching assistants if discourse continues. Beyond conflict, we will be sure to ask questions of professors and TA's with any uncertainties that arise.

8. Goals and Objectives
A successful project looks like a dashboard that succinctly provides information on the risk of fires in scale, spread, and destruction. We would like to include a selection of plots that shows the relationship between fire variables, as well as maps that show spatial relationships of fire in California. To check the success of our dashboard, we will test the dashboard as if we are a non-academic resident of California looking to gain information on potential risks of fire in our areas. Essential deliverables include plots showing analysis of size, speed, and destruction of fires versus potential independent variables including cause, location, or other spatial data. Optional features could include a model to predict the potential spread of fire or more interactive components of the dashboard.

9. Schedule and Milestones
Work backward from the assignment deadlines to establish internal milestones for:

Defining research questions and project scope: October 15-22 **complete**
Assessing data feasibility: October 15-22 **complete**
Preparing and combining datasets: October 15-29
Conducting exploratory analysis: October 29-November 5
Building a dashboard prototype: November 5-12
Reviewing and testing the dashboard: November 11-13
Preparing the final presentation: November 13-18
Final project presentation: November 19

We will continue to update these and assign work week by week as additional commitments arise. We are all responsible for the successful completion of every milestone. We will track progress in class and our weekly meetings, and we will revisit the scope at each step, particularly following the exploratory analysis and as we begin to build the dashboard prototype.